Parktown Boys' High School

District: 9

Geography Department

JUNE EXAM

PAPER 1

Grade 12

MEMO

Examiner: Ms A Clayton	Time: 3 Hour
Moderator: Mr A Meintjes	Total: 150 Marks
Number of pages: 17	Date: 1 June 2026

SECTION A: CLIMATE AND WEATHER AND GEOMORPHOLOGY

QUESTION 1: CLIMATE AND WEATHER

1.1.

1.1.1	**south**
1.1.2	**South Indian**
1.1.3	**ridge**
1.1.4	**1 016 hPa**
1.1.5	**10 knots**
1.1.6	**north-west**
1.1.7	**subtropical high**

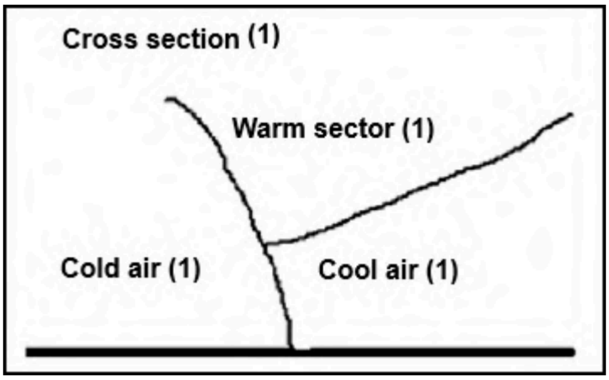
(7x1) (7)

1.2

1.2.1	**C**
1.2.2	**B**
1.2.3	**C**
1.2.4	**A**
1.2.5	**B**
1.2.6	**C**
1.2.7	**B**
1.2.8	**D**

(8 x 1) (8)

1.3 Refer to the synoptic weather map below on mid-latitude cyclones.

1.3.1	What do we call a series of mid-latitude cyclones in succession? (1 x 1) (1) **Family (of cyclones) (1)**
1.3.2	Name front A and sector B on the synoptic weather map. (2 x 1) (2) **A – Cold front (1)** **B – Warm sector (1)**
1.3.3	What influence does the South Indian Anticyclone, as shown on the synoptic weather map, have on the path followed by the mid-latitude cyclones? (1 x 2) (2) **Blocks /blocking HP their path to the east. (2)** **Causes them to change their path to a south-easterly direction. (2)** **[Any ONE]**
1.3.4	The mid-latitude cyclone at C is in the cold occlusion stage. Draw a labelled cross-section at C on the synoptic weather map. Marks will be awarded for the following: • **1 mark for the correct drawing of the Cross-section – COLD FRONT LINE MUST TOUCH THE SURFACE (1)** • **1 mark for the indication of the warm sector (1)** • **1 mark for the indication of the location of the cold air (1)** • **1 mark for the indication for the location of the cool air (1)** 
1.3.5	Describe the changes in weather that Cape Town will experience as the mid-latitude cyclone passes over. (3 x 2) (6) **NB: The student must describe the change (e.g. increasing, decreasing, etc.) and the statement must be developed/expanded point. They cannot be credited for simply stating or listing the change, e.g. "The temperatures decrease. Pressure increases. Rainfall Increases" = 0** **Decrease in temperatures because of cold air behind the cold front/ Temperature decrease as warm air is displaced by cold air (2)** **Wind direction changes (backs) from north-westerly before the front to south-westerly after the front has passed. (2)** **Wind speeds will increase/become stronger due to a steep pressure gradient/strong upliftment of warm air (2)** **Heavy rainfall occurs due to steep uplift of the air at the front. (2)** **Air pressure initially decreases as air rises and then increases after the front (2)** **[ANY THREE]**

1.4 Refer to the figure below which shows methods to reduce the effects of a heat island.

1.4.1	Define the term urban heat island. **It is an urban area of higher temperature surrounded by rural areas of lower temperature. (2)** **[CONCEPT]** (1 x 2) (2)
1.4.2	Which part of an urban area is likely to experience the highest temperatures? Choose the correct answer and write only the letter. **B** (1 x 1) (1)
1.4.3	Describe how the glass windows in the urban area shown in the figure will result in the increase of temperature in the urban area. **It lets heat in and acts as a greenhouse that traps the heat inside the building. (2)** **It reflects heat between buildings. (2)** **[Any ONE]** (1 x 2) (2)
1.4.4	Explain TWO possible negative effects that urban heat islands have on the health of human beings. **It increases human discomfort because of high temperatures. (2)** **Heat stress/stroke/cardiovascular illnesses during heat waves (2)** **Urban smog reduces visibility and may cause accidents. (2)** **Loss of energy to be productive in working environment(2)** **Increases the number of insects such as mosquitoes and fleas which infect/irritate humans (2)** **[Any TWO – must be a negative effect on humans]** ***Note: UHI effect does not increase UV radiation and cause more skin cancer.*** (2 x 2) (4)

1.4.5	Explain how the THREE strategies (A, B and C) evident in the figure contribute to the reduction of the heat island effect. ***Note: Strategies must explain how to reduce the UHI Effect, not reduce the effects of the UHI. E.g. for C water for people to drink cannot be accepted as it does not reduce the UHI effect, it reduces the effect of the UHI.*** **A – By planting more trees, it reduces the amount of heat through absorption. (2)** **Transpiration by trees lowers temperatures. (2)** **Shadows of the trees will cool the temperatures. (2)** **Reduces CO2 which is a GHG that absorbs heat. (2)** **[Any ONE]** **B – Dense buildings prevent air from moving freely between buildings, causing heat to build up. (2)** **Space between buildings lowers temperatures because air moves freely and winds increase. (2)** **Cooler winds increase between buildings with significant space. (2)** **[Any ONE]** **C – Water sources absorb heat and use it in evaporation. (2)** **Evaporation of water uses heat and lowers temperatures. (2)** **Water is cooler than surfaces such as sand, tar, concrete etc. (2)** **[Any ONE]** **[Must refer to A, B and C. All THREE must be explained.]** (3 x 2) (6)

1.5 Refer to the figure below which should a line thunderstorm.

1.5.1	On which side of the moisture front do line thunderstorms occur? East of west? **East (1)** **Note: Not right. Must give a direction.** (1 x 1) (1)
1.5.2	Give a reason for your answer to QUESTION 1.5.1. **The cold air from the west uplifts the warm, moist air on the east of the moisture front. (Rising air cools and condenses and forms clouds.) (2)** (1 x 2) (2)
1.5.3	Describe the air that flows from the: ***Note: Must give two adjectives for each response to score the 2 marks*** a. South Indian Anticyclone (1 x 2) (2) **The warmer, moister and less dense air (2)** b. South Atlantic Anticyclone (1 x 2) (2) **Colder, drier and denser air (2)**
1.5.4	In a paragraph of approximately EIGHT lines, suggest the positive and negative impact that line thunderstorms have on the agricultural sector in the interior of South Africa. **POSITIVE:** **Line thunderstorms bring rain to irrigate crops. (2)** **Line thunderstorms bring rain to irrigate pasture land for livestock. (2)** **It can fill dams creating availability of water during dry periods (2)** **It replenishes the ground water as well as rivers to be used for crops/cattle. (2)** **Gives relief to droughts or semi-arid areas practicing agriculture. (2)** **NEGATIVE:** **Thunderstorms can damage/flood crops. (2)** **Thunderstorms can injure/kill/drown cattle. (2)** **The gusty wind, hail or rain can wash away/erode the fertile soil. (2)** **Affect trading/production of food that impacts on the agricultural sector (2)** **Damage to agricultural infrastructure due to flooding (2)** **Silt is washed into dams, reducing their water holding capacity which is much needed for crops/livestock. (2)** **[Any FOUR facts. Must include both positive and negative impacts]** (4 x 2) (8)

[Question 1: 60 marks]

QUESTION 2: GEOMORPHOLOGY

2.1.

2.2.1	**E**
2.2.2	**G**
2.2.3	**C**
2.2.4	**F**
2.2.5	**D**
2.2.6	**H**
2.2.7	**B**

(1 x 7) (7)

2.2

2.2.1	**B**
2.2.2	**G**
2.2.3	**A**
2.2.4	**C**
2.2.5	**E**
2.2.6	**H**
2.2.7	**D**
2.2.8	**F**

(1 x 8) (8)

2.3 Refer to the figure below depicting too different drainage patterns.

2.3.1	Identify drainage patterns A and B. **A Trellis (1)** **B Dendritic (1)**	(2 x 1) (2)
2.3.2	Differentiate between the underlying rock structure of drainage patterns A and B respectively. **A Alternate layers/bands of hard and soft rock/ folded rock structure (2)** **B Rock that is uniformly resistant to erosion (2)**	(2 x 2) (4)
2.3.3	Why are the tributaries of the main stream parallel to each other in drainage pattern A? **The streams flow in relation to the folds of the rock (2)** **The streams flow over softer rock of the syncline (valley) (2)** **Interfluves are parallel (2)** **[ANY ONE]**	(1 x 2) (2)
2.3.4	Determine the stream order at point 1 in drainage pattern B. **3 rd (order) (2)**	(1 x 2) (2)
2.3.5	Choose the CORRECT word between brackets to make the statement TRUE. The higher the stream order, the (higher/lower) the drainage density. **Higher (1)**	(1 x 1) (1)
2.3.6	Describe the relationship between: a. Drainage density and low rainfall (1 x 2) (2) **The low rainfall will result in a lower drainage density (2)** b. Drainage density and steep gradient (1 x 2) (2) **The steep gradient will result in a higher drainage density (2)**	

2.4 Refer to the sketch map of rivers Y and Z before river capture has taken place.

2.4.1	Define the concept river capture. (1 x 2) (2) **When a more energetic river captures the headwaters of a less energetic river (2)**
2.4.2	State ONE condition needed for river capture to take place. (1 x 2) (2) **A steeper gradient (on the one side of the watershed) (2)** **More rainfall (on one side of the watershed) (2)** **Less resistant/softer rock (on the one side of the watershed) (2)** **Headward erosion of the captor stream (2)** **[ANY ONE]**

2.4.3	Draw a sketch to illustrate the area after river capture has taken place. (1 + 3) (4)
	Marks allocated as follows:
	• **Accuracy of sketch- any one of two tributaries can be used (1)**
	• **Labels:**
	    ○ **Wind gap (1)**
	    ○ **Elbow of capture (1)**
	    ○ **Misfit stream (1)**

2.4.4	Will river Y or Z experience rejuvenation after river capture? (1 x 1) (1) **Y**
2.4.5	Give a reason for your answer to QUESTION 2.4.4. (1 x 2) (2) **River Y has an increased volume of water/increased discharge (2)**
2.4.6	Refer to your answer to QUESTION 2.4.5 and explain the impact of the change on the captor stream. (2 x 2) (4) **Note: This question refers to the impact on the captor stream, it doesn't mention social/economic or environmental impact. Credit impacts on the stream only.** **Increased vertical erosion due to the increased volume of water in river Y (2)** **The active erosion of the river cuts into the valley forming terraces (2)** **The softer rock in the valley erodes faster resulting in layers/terraces (2)** **New valleys form in a valley due to increased river discharge (2)** **Terraces form due to recurrent rejuvenation in several valleys (2)** **Meanders will become incised/entrenched (2)** **A knickpoint can develop along the profile of the river (2)** **Increased flooding because of greater volume of water (2)** **Increased velocity of water in the river channel because of greater volume of water (2)** **The captor stream will be able to carry a greater load/less deposition (2)** **[ANY TWO]**

2.5 Study the extract on Gauteng River Catchment Management below.

2.5.1	Define the concept river management. **The planned and coordinated use of the river without compromising people's well-being and the environment (2)** **[CONCEPT]** (1 x 2) (2)
2.5.2	Give ONE piece of evidence from the extract that shows the importance of catchment management in Gauteng. **Healthy riverbanks maintain the form of the river channel. (1)** **It provides habitat for species (aquatic and terrestrial) and filter sediment, minerals and light. (1)** **Water quality includes the chemical; physical and bacteriological properties of water that determines its suitability for use. (1)** **[Any TWO]** (1 x 1) (1)
2.5.3	Discuss TWO economic impacts that that poor river management would have for Gauteng. **Note: Response must have a factor and qualifier. The must be a link to an economic impact to be credited.** **Poor river management can lead to water pollution and reduced water availability, which can <u>reduce agricultural productivity</u> and farmers' income.** **Poor river management can increase water pollution, which can <u>raise the cost of water treatment</u> for businesses and municipalities.** **Poor river management can result in polluted and degraded river environments, which can <u>reduce tourism revenue</u>.** **Poor river management can increase the risk of flooding, which can damage infrastructure and <u>lead to costly repairs</u>.** **Poor river management can cause water shortages and poor water quality, which can <u>reduce industrial production and profits</u>.** **Poor river management can negatively affect economic sectors such as agriculture, tourism, and industry, which can lead to <u>job losses and higher unemployment</u>.** **[Any TWO]** (2 x 2) (4)

2.5.4	In a paragraph of approximately EIGHT lines, provide sustainable strategies that can be implemented to deal with problems associated with poor river management systems. **Strict local by-laws on disposal of waste from both domestic use and industry (2)** **Buffering the area to prevent pollution (can give examples) (2)** **Maintaining the natural vegetation and reducing deforestation (2)** **Removing Alien invasive species (2)** **Frequent monitoring of water quality (2)** **Education and public awareness on environmental management (2)** **Regular cleaning of riverbanks and surrounding areas (2)** **Implementing of fines for violation of environmental regulations (2)** **[Any FOUR]** (4 x 2) (8)

[Question 2: 60 marks]

Section A Total: 120 marks

SECTION B

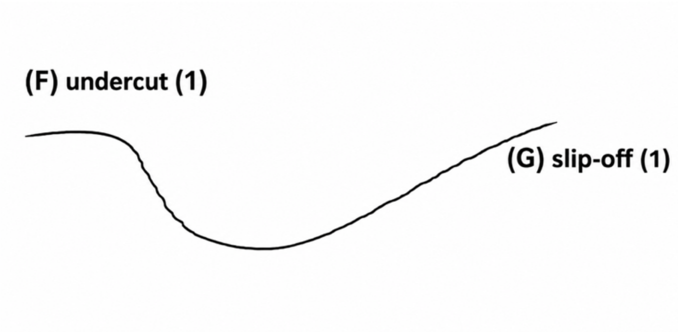
QUESTION 3: GEOGRAPHICAL SKILLS AND TECHNIQUES

3.1 MAP SKILLS AND CALCULATIONS

Refer to the topographical map and the orthophoto map.

3.1.1	In which province is Malelane located? **C** (1 x 1) (1)
3.1.2	The true bearing from spot height 557 in E5 to spot height 522 in E4 on the topographic map is: **C (error with options on the QP. 242 is the actual answer, C is the closest)** (1 x 1) (1)
3.1.3	Using your answer in QUESTION 3.1.2, calculated magnetic bearing. **Answer from 3.1.2 (242/282)+ 18°28' (1)** **Answer: 260/300°28' (1)** (2 x 1) (2)
3.1.4	Refer to the large perennial water body (dam) in E4 on the topographic map. Calculate the area covered by the dam in D1 if the length is 1cm and the breadth is 0,8cm. Give the answer in km². Show ALL calculations. **1cm x 0,5 = 0,5km (1) (Conversion of length)** **0,8 x 0,5 = 0,4km (1) (Conversion of breadth)** **0,5 x 0,4 = 0,2km² (1) (answer in km²)** (3 x 1) (3)
3.1.5	Choose the correct answer from the options provided below. Write only the letter (A – D) next to the question number. A cross section of the slope between **8** in block **D4** and **9** in block **D3** on the orthophoto map extract is .... **B** (1 x 1) (1)
3.1.6	Give a reason for your answer to QUESTION 3.1.5. **Contour lines are steeper at the top (at 8) and more gentle at the bottom (at 9) (2)** **The contour lines are far apart at a high lying area and closer together at a low lying area (2)** **[ANY ONE]** (1 x 2) (2)

3.2 MAP INTERPRETATON

3.2.1	Much of the agricultural area surrounding Malelane is in the valley of the Umgwenya river.
	(a) Choose the correct answer from the options provided below. Write only the letter (A – D) next to the question number. The nocturnal (night-time) wind that develops during the night in this area is called a ... wind. **D** (1 x 1) (1)
	(b) Explain how the wind identified in QUESTION 3.2.1(a) promotes the formation of dense fog at **K**. **Valley slopes cool at night and air sinks to the bottom of the valley. (2)** **Decrease in temperature results in condensation taking place close to the surface in the morning. (2)** **[Any ONE]** (1 x 2) (2)
	(c) Give TWO impacts that the development of fog in this area could have on early morning traffic on the national route near **K**. **It will result in poor visibility. (1)** **It will slow down traffic./lengthen travel time (1)** **Might cause accidents (1)** **[Any TWO]** (2 x 1) (2)
3.2.2	Draw a cross-section along the line **F** - **G** across the Umgwenya River on the topographic map. Award marks: • **Asymmetrical cross section from F to G, with side at F deeper than side at G (1)** • **Label of undercut (1)** • **Label at slip-off slope (1)**  (1 + 2 x 1) (3)

3.2.3	Refer to point J on the topographic map. Explain why deposition is the main process taking place at this point. **Slow moving water does not have the carrying capacity. (2)** **River velocity is low and larger fluvial particles are deposited. (2)** **On the inside of the bend, where the river flow is slower, material is deposited, as there is more friction. (2)** **[ANY ONE]** (1 x 2) (2)
3.2.4	Refer to block **C2** on the topographical map.
	(a) Identify the fluvial feature at **H** in block **C2** on the topographical map. **Braided stream (1)** **Eyot (1)** **[ANY ONE]** (1 x 1) (1)
	(b) The feature identified in QUESTION 3.2.3 (a) is commonly found in the ... of a river where the river valley has a/an ... **C** (1 x 1) (1)

3.3 GEOGRAPHIC INFORMSTION SYSTEMS (GIS)

3.3.1	Refer to the dam in block A5 on the topographical map.
	(a) Is the dam an example of (vector/raster) data? (1 x 1) (1) **Vector (1)**
	(b) Give a reason for the answer to QUESTION 3.3.1 (a). (1 x 2) (2) **Vector data is given in lines, points/dots and polygons (2)**
3.3.2	(a) Define the term buffering. (1 x 2) (2) **Buffering is a zone that is drawn around any point, line, or polygon that encompasses all of the area within a specified distance of a feature e.g., river. (2)** **[CONCEPT]**
	(b) Do you think buffering was applied in the development of the house within Leopard Creek Golf Estate in block **C2**? (1 x 1) (1) **Note: Read the answer to 3.3.2 (c). If the application or lack of application of buffering is valid, then credit yes or no accordingly.** **No (1) / Yes (1)**
	(c) Give a reason from the topographic map for your answer to QUESTION 3.32 (c). (1 x 2) (2) **No- Development of the houses is against a riverbank with no indication of a buffer area on the side of the river. (2)** **Yes – The woodlands are a buffer between the houses/buildings (2)**

[Question 3: 60 marks]

Section B Total: 30 marks

Exam Total: 150 marks